Data quality report

Maple Bank collections release → Kept's golden C360 · deliverable 4

Team Sharpe Shooters

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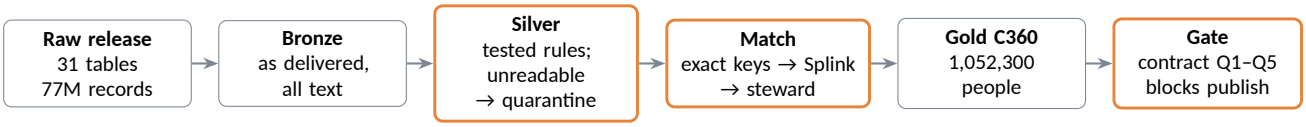
TL;DR

We took **31 tables / 68,162,322 records** of deliberately messy bank data and turned them into one trusted record per person. Every number here is a live query (`scripts/quality_report_full.py`); nothing's typed in.

Quality gate on the golden record	**PASS, all 8 rules**
Source keys → people	**3,840,513 → 1,052,300 golden IDs** (biggest person: 12 keys)
Match precision, checked by a human	**99.3%** (298/300; exact 140/140, Splink 158/160)
Kinds of data issue found and handled	**19** (worst: autopay / pre-authorised-debit account at another bank (not one of our deposit accounts), 64.70% of its table)
Clusters we refused to auto-merge	5,491 (two national IDs) + 7 (more than 12 keys)
Training vs live feature values	PASS: 0 of 56,000 cells differ (0.0000%; the limit is 0.1%).
PII left in 785,594 redacted notes and calls	**0**

The vibe: never guess. If a rule can't read a value, it goes empty and into a quarantine table with the reason, instead of being "fixed" into something wrong. A wrong date of birth can merge two strangers into one customer, and in collections that means calling the wrong person about someone else's debt.

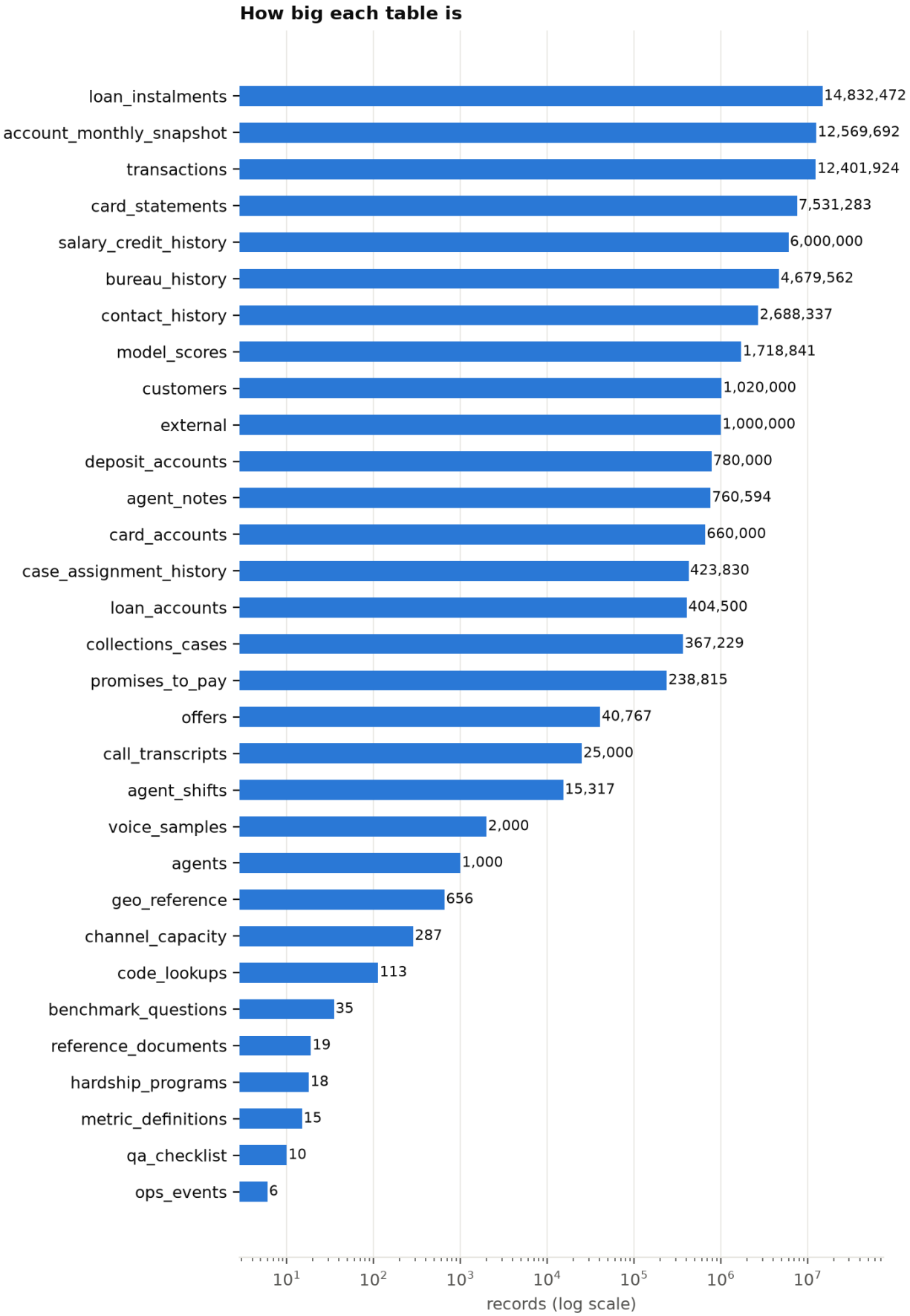
1. How we checked



checks at every step: dbt tests · Soda checks from the contract · the gate · a human for the 300-pair review and the steward queue

- **Land raw, clean after.** Bronze keeps every file exactly as it came, all columns as text, so nothing gets silently converted (the README warns `NA` is a real outcome code and IDs have leading zeros; both survive).
- **Small tested rules** in `kept.data.clean` / `kept.data.traps` do the cleaning: names, dates, phones, postal codes, provinces, snapshots, the renamed column.
- **Four layers of checks:** dbt tests on every model (gold types enforced), Soda checks generated from the data contract, the contract gate (`kept gate`, blocks publishing), and a human for what code can't decide (300 sampled matches, the steward queue).

2. What’s in the release



Missing values are mostly by design: a charge-off date only exists if the account was charged off, a reversal ID only if the transaction was reversed. So high numbers here aren't bugs; §3 is where the real problems are.

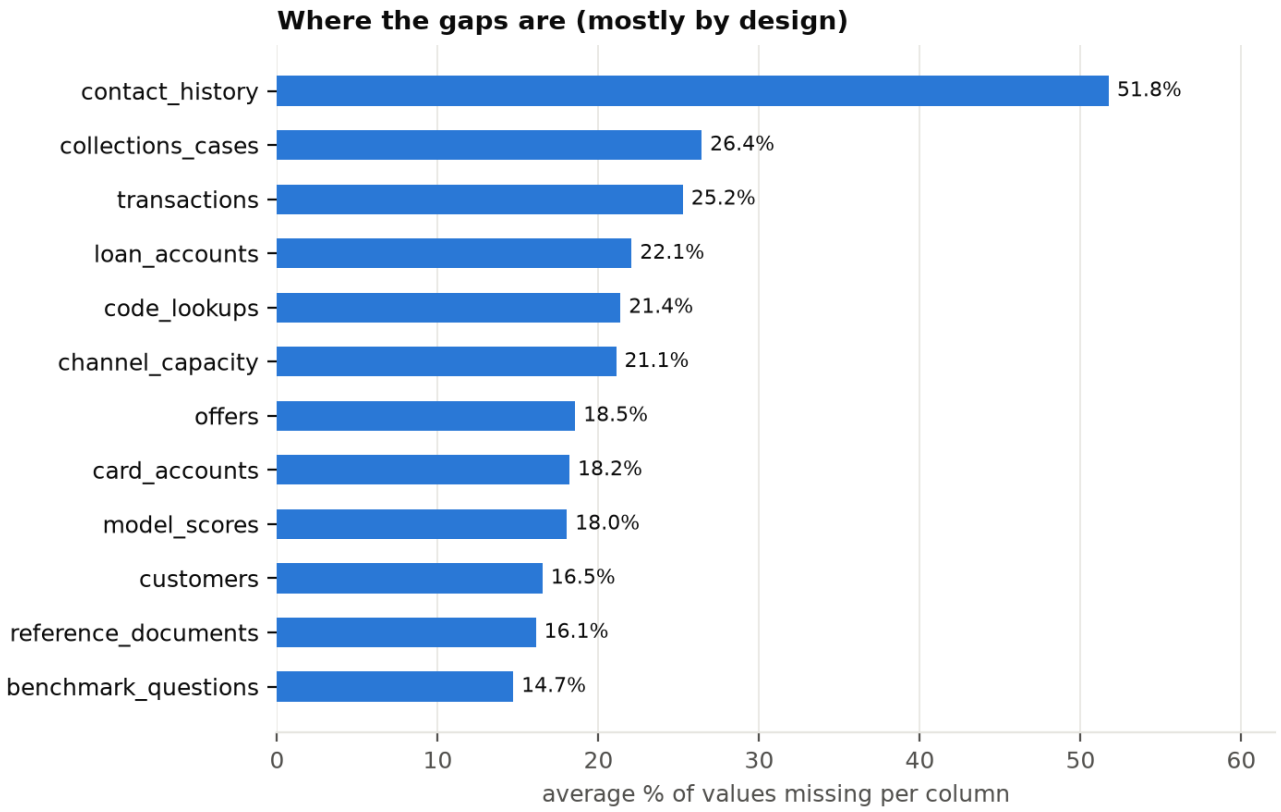
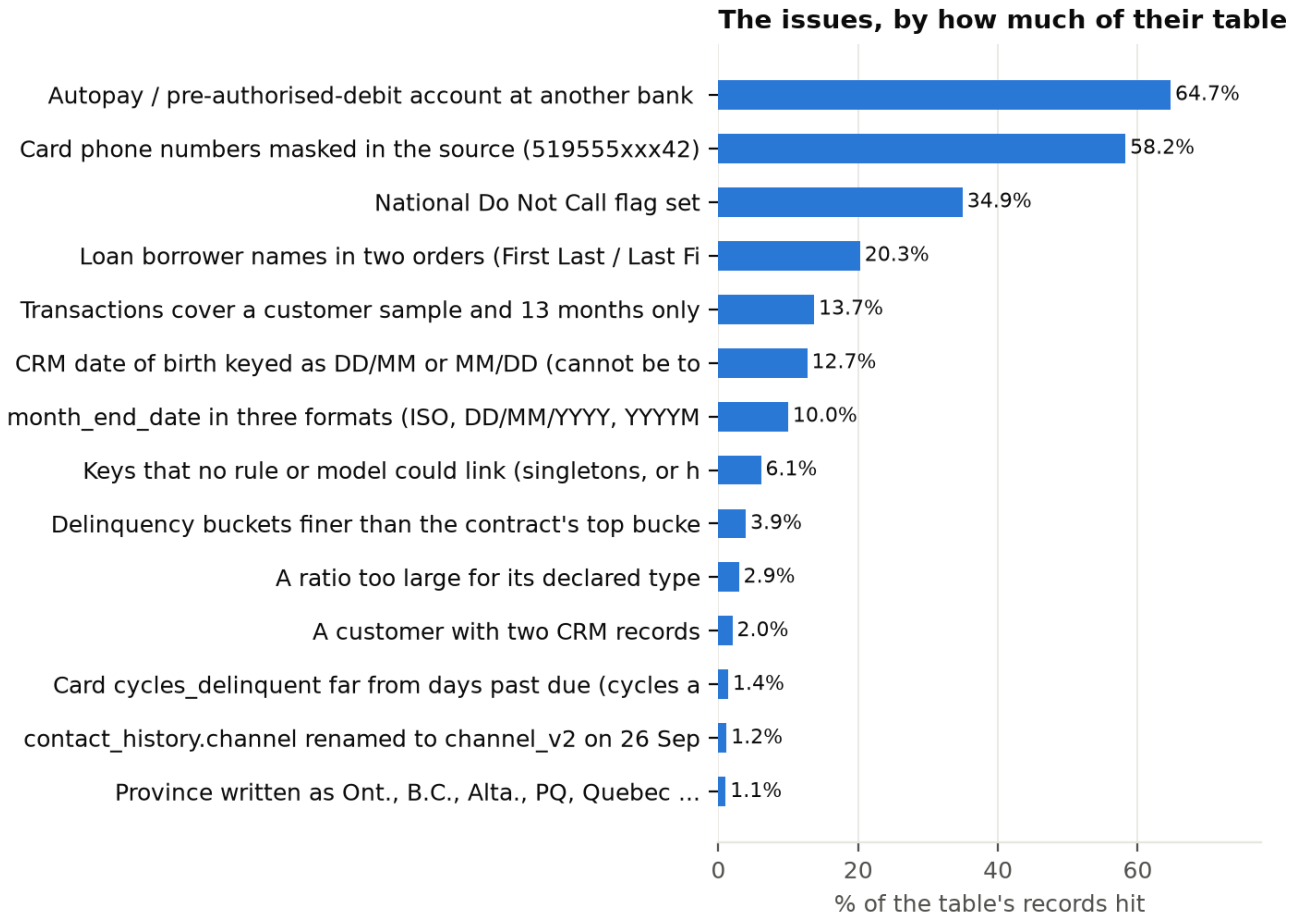


Table	Rows	Cols	Cols with gaps	Avg % missing	Emptiest columns
loan_instalments	14,832,472	19	5	13.5	days_late_when_paid, pad_retry_date
account_monthly_snapshot	12,569,692	16	5	13.1	salary_credit_amount, limit_or_original_amount
transactions	12,401,924	25	10	25.2	reversal_of_txn_id, return_reason
card_statements	7,531,283	26	3	7.8	days_late, first_payment_after_due_date
salary_credit_history	6,000,000	11	2	5.5	first_credit_date, last_credit_date
bureau_history	4,679,562	10	1	0.5	risk_score
contact_history	2,688,337	128	90	51.8	record_hash, voice_sample_id
model_scores	1,718,841	33	15	18.0	predicted_value, ptp_id
customers	1,020,000	148	66	16.5	record_hash, duplicate_of_crm_id
external	1,000,000	122	32	10.3	public_record_date, risk_score_m01
deposit_accounts	780,000	145	19	8.3	freeze_reason, overdraft_dpd
agent_notes	760,594	26	6	8.9	last_edited_ts, template_id

Table	Rows	Cols	Cols with gaps	Avg % missing	Emptiest columns
card_accounts	660,000	148	75	18.2	chargeoff_date, chargeoff_amount
case_assignment_history	423,830	23	2	2.0	routing_rule_id, unassigned_ts
loan_accounts	404,500	135	76	22.1	chargeoff_date, chargeoff_amount
collections_cases	367,229	132	59	26.4	record_hash, agency_name
promises_to_pay	238,815	29	7	8.0	broken_reason_agent_coded, paid_date
offers	40,767	19	8	18.5	decline_reason, plan_start_date
call_transcripts	25,000	29	1	3.2	voice_sample_id
agent_shifts	15,317	15	2	9.5	absence_reason, worked_minutes
voice_samples	2,000	34	1	0.9	channel_map
agents	1,000	30	0	0.0	-
geo_reference	656	15	1	1.0	cma_name
channel_capacity	287	11	4	21.1	ops_event_id, daily_budget_cad
code_lookups	113	7	2	21.4	description_fr, attributes_json
benchmark_questions	35	7	2	14.7	metric_id, tolerance
reference_documents	19	17	3	16.1	related_table, superseded_by
hardship_programs	18	21	3	8.5	term_extension_max_months, rate_override
metric_definitions	15	14	0	0.0	-
qa_checklist	10	10	0	0.0	-
ops_events	6	10	1	3.3	reference_doc_id

3. The issues, and what we did about each

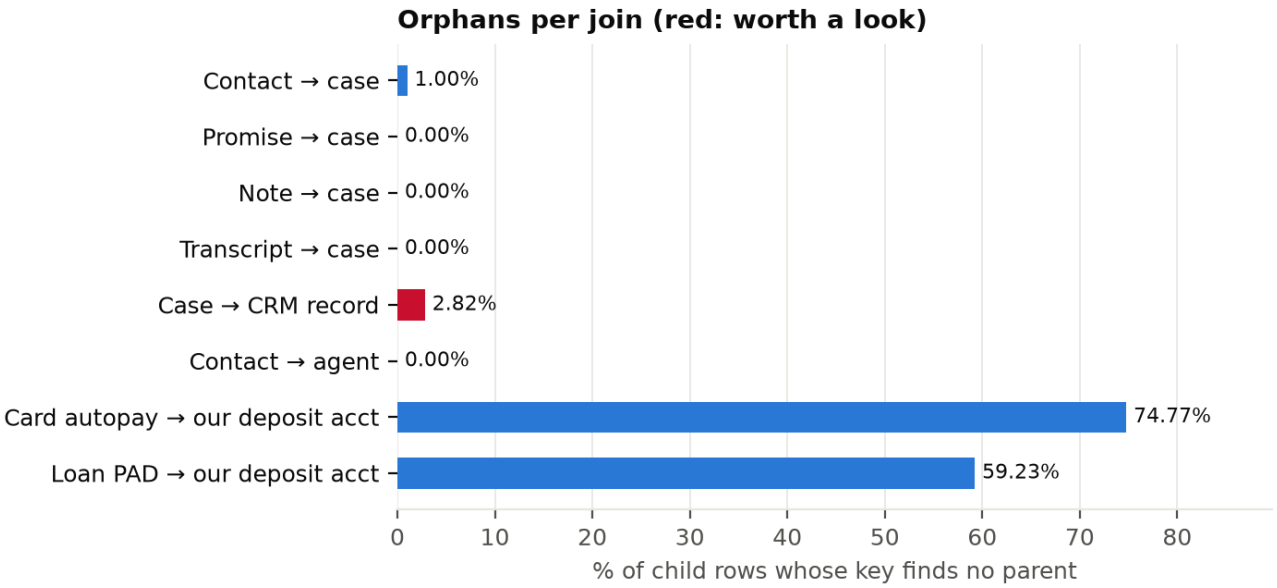


Issue	Where	Records hit	Fix
Agent notes with lost characters (U+FFFD)	`agent_notes.note_text`	4,636 (0.61%)	Not recoverable: the original characters are gone. Left as is; the redactor and extractor read the surrounding words, which are intact.
CRM records deleted upstream (CDC) but still in the extract	`customers.is_deleted`	2,115 (0.21%)	Left out of silver (the current version is deleted); listed in quarantine.customers.
A customer with two CRM records	`customers (crm_customer_id)`	20,434 (2.01%)	Linked by the same national ID, CRM's own duplicate pointer, or Splink (name, DOB, phone, postal code); one golden ID each. The count is CRM records beyond the first of each golden ID.
Card phone numbers masked in the source (519555xxx42)	`card_accounts.cardholder_phone_raw`	384,422 (58.25%)	Cannot be normalised to a phone number: left null in the cleaned column and listed in quarantine.card_accounts. Cards link on name, DOB and postal code instead.
CRM phone numbers cut to 9 digits	`customers.primary_phone_raw`	4,936 (0.48%)	Not a valid number: left null in the cleaned column, listed in quarantine.customers.

Issue	Where	Records hit	Fix
CRM date of birth keyed as DD/MM or MM/DD (cannot be told apart)	`customers.dob_raw`	128,980 (12.71%)	Never guessed (reading every slash date day-first misreads 16% of them). CRM also holds a clean date_of_birth, which is used; the raw value only fills gaps when it is unambiguous. Cards, whose format is fixed DD-MM-YYYY, are read day-first.
month_end_date in three formats (ISO, DD/MM/YYYY, YYYYMMDD)	`account_monthly_snapshot.month_end_date`	1,255,152 (9.99%)	A plain date cast silently dropped these (1.25M values). Silver reads all three; a month-end day is always above 12, so the DD/MM form is unambiguous.
Province written as Ont., B.C., Alta., PQ, Quebec …	`customers.province_code`	10,689 (1.05%)	Mapped to the two-letter code (normalize_province); gold would otherwise break the contract.
A ratio too large for its declared type	`deposit_accounts.income_to_debt_service_ratio_est`	13,969 (2.94%)	The data dictionary says decimal(8,4); values go above 9999.9999. Silver uses a wider decimal. The magnitudes look like a unit problem upstream: not used by any feature.
contact_history.channel renamed to channel_v2 on 26 Sep	`contact_history.channel, channel_v2`	31,428 (1.17%)	Silver coalesces the two into channel (the dictionary name).
Loan borrower names in two orders (First Last / Last First)	`loan_accounts.borrower_name_raw`	82,028 (20.28%)	The token CRM uses more as a surname is the surname (silver.int_name_lexicon). Without it, loan names matched other systems 79% of the time; with it, 98%.
Autopay / pre-authorised-debit account at another bank (not one of our deposit accounts)	`card_accounts.autopay_source_account, loan_accounts.pad_source_account`	302,549 (64.70%)	Ignored for linking: there is nothing at our bank to link to. Left untouched in silver.
National Do Not Call flag set	`customers.national_dncl_flag`	355,697 (34.94%)	Never blocks collections contact (the list covers telemarketing, not collections). Internal do-not-call and cease flags do.
Delinquency buckets finer than the contract's top bucket (121-150, 151-180, 180+)	`collections_cases.current_bucket`	14,409 (3.92%)	Silver keeps the source label. Gold derives its bucket from days past due, with one 121+ bucket (the shared contract).
Card cycles_delinquent far from days past due (cycles are not days)	`card_accounts.cycles_delinquent, dpd`	673 (1.41%)	Gold and features use dpd, never cycles_delinquent.
Contact rows whose case is not in collections_cases (orphans; the example contract allows < 1%)	`contact_history.case_id`	26,864 (1.00%)	Counted and kept in silver; nothing joins them to a case.
Duplicate primary keys in the source tables	`customers, contact_history, promises_to_pay, agent_notes`	0 (0.00%)	None found when this report was written (the count above is the proof).
Transactions cover a customer sample and 13 months only	`transactions`	106,914 (13.71%)	Missing stays missing: features from transactions are null, not 0, outside the sample.
Keys that no rule or model could link (singletons, or held apart)	`gold.c360_customer`	64,092 (6.09%)	Each keeps its own golden ID with status 'unlinked'. 5,498 clusters were held (two national IDs, or more than 12 keys) and go to the steward queue.

Schema drift. `kept map` diffs every raw table's columns against the data dictionary. It flags `deposit_accounts.overdraft_dpd / overdraft_dpd_bucket` (the README mentions them, the dictionary sheet doesn't), `benchmark_questions`, and the transcript turn fields (they live in the JSON, not the CSV index). `contact_history.channel` → `channel_v2` (renamed 26 Sep) gets merged back in silver.

4. Do the tables actually join?



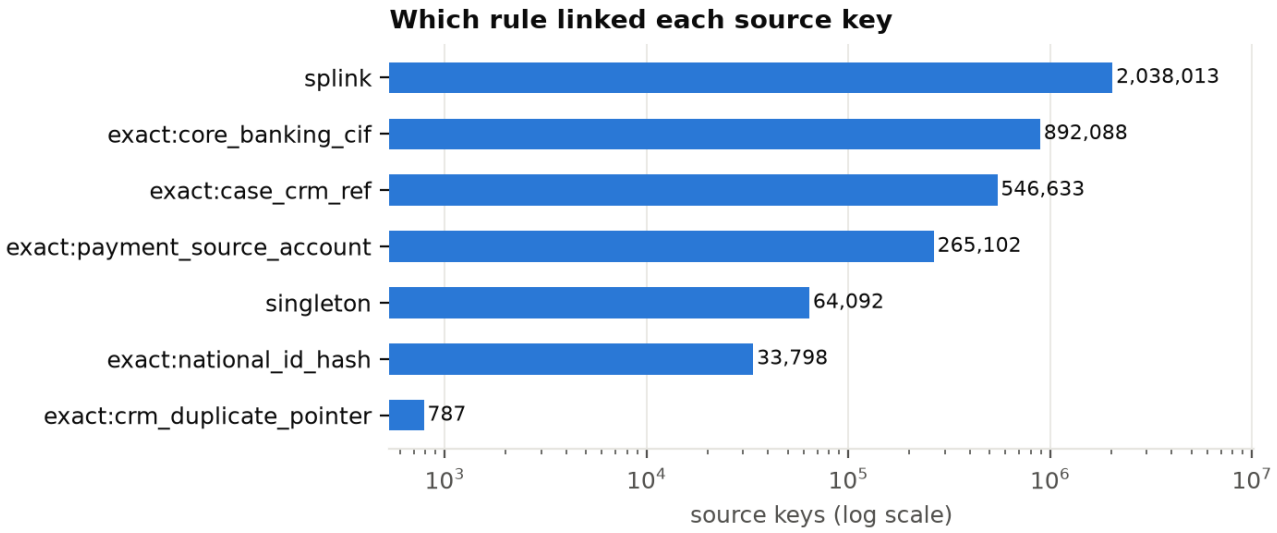
Join	Child rows with a key	Orphans	Share	What happens
Contact → case	2,688,337	26,864	1.00%	kept and counted; nothing joins them to a case
Promise → case	238,815	0	0.00%	promise features join on case_id
Note → case	760,594	0	0.00%	notes still feed hardship by customer
Transcript → case	25,000	0	0.00%	call summaries look calls up by case
Case → CRM record	367,229	10,365	2.82%	other exact rules + Splink still resolve them; every open case resolves (gate Q2)
Contact → agent	1,018,776	0	0.00%	blank for automated contacts
Card autopay → our deposit acct	164,565	123,047	74.77%	account at another bank: ignored for matching
Loan PAD → our deposit acct	303,037	179,502	59.23%	same

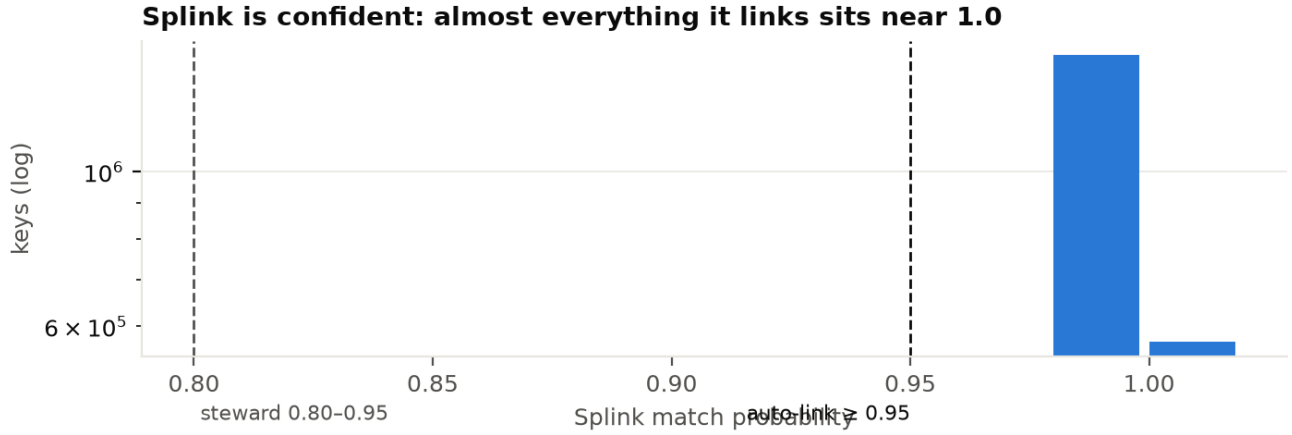
The one worth flagging: **10,365 cases (2.82%) point at a CRM ID that isn't in `customers`.** They're not lost: card, loan and deposit keys plus Splink still put them on a person, and the gate checks that every *open* case resolves. The autopay/PAD numbers are high on purpose: most of those accounts are at other banks, so we never use them to link anyone.

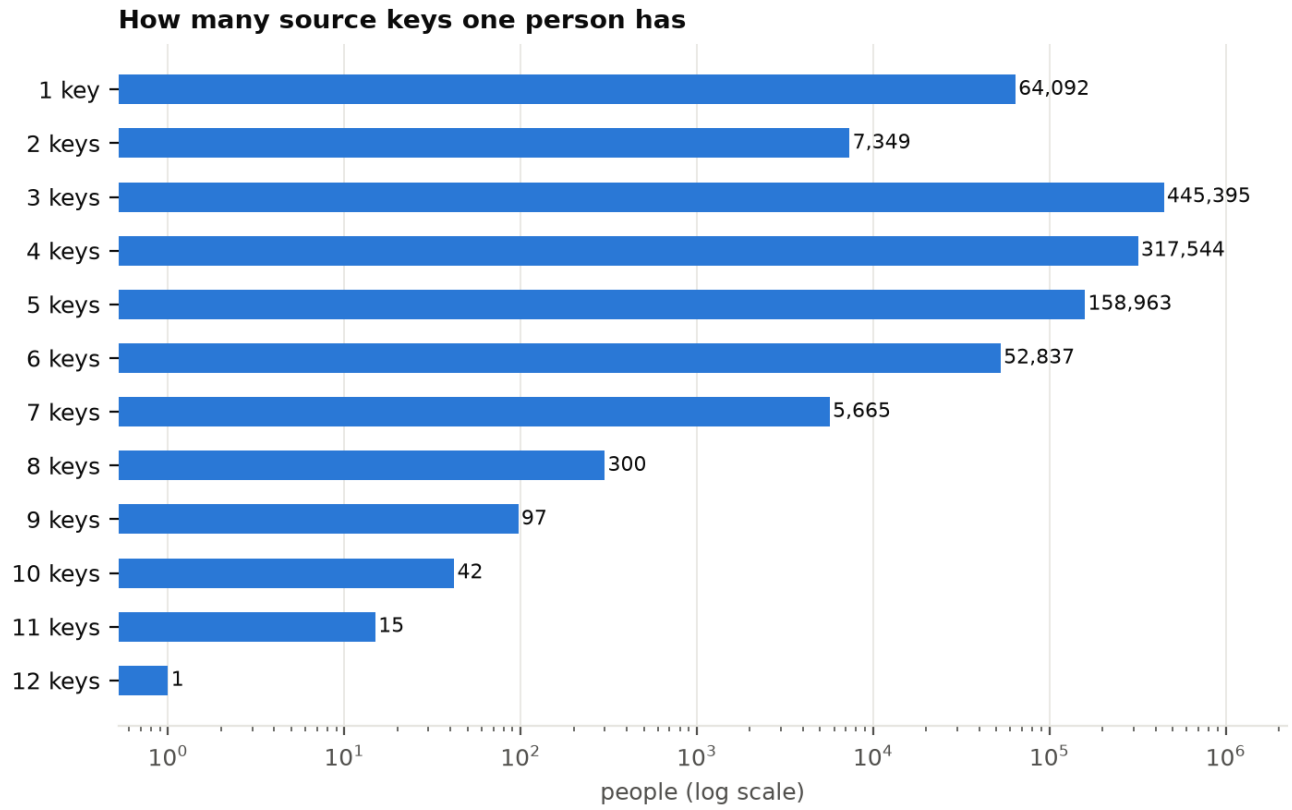
5. Identity resolution (the bit that can actually hurt someone)

There's no shared customer ID across the five systems. We link in two passes:

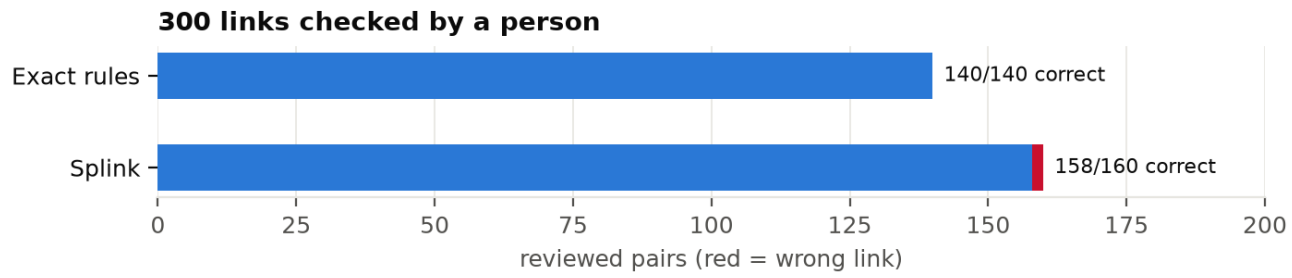
1. **Exact rules**: keys we can prove (the case's copy of the CRM ID, autopay/PAD → our deposit account, the same national ID hash, CRM's own duplicate pointer, and the core-banking customer number shared by deposits and the bureau file).
2. **Splink** (Fellegi–Sunter) on what's left, comparing name, date of birth, phone and postal code. **≥ 0.95 links automatically, 0.80–0.95 goes to a data steward, lower stays apart.** A cluster with two national IDs or more than 12 keys is never auto-merged.


Which rule linked each source key
splink
exact:core_banking_cif
exact:case_crm_ref
exact:payment_source_account
singleton
exact:national_id_hash
exact:crm_duplicate_pointer
2,038,013
892,088
546,633
265,102
64,092
33,798
787
10^3
10^4
10^5
10^6
10^7
source keys (log scale)


Splink is confident: almost everything it links sits near 1.0
keys (log)
10^6
6×10^5
0.80
0.85
0.90
0.95
1.00
steward 0.80–0.95
auto-link ≥ 0.95
Splink match probability



The human check. No ground-truth IDs came with the data, so a person reviewed 300 random predicted links, sorted into tiers by how many identifiers agree (tier 1: 276, tier 2: 14, tier 3: 10).



Both wrong links (MP-238, MP-239) are Splink pairs where everything matched except the surname, and both scored 1, 1. A stricter threshold wouldn't have caught them; the human did. That's why the review sample stays in the pipeline.

Golden IDs are stable run to run (last run: 0 merged, 0 split). The golden record has 1,052,300 people: 64,092 are single keys nothing could link, 37,042 carry `do_not_contact` and 23,593 `collections_hold`, each set if **any** linked record says so. Hardship status: none 991,933, confirmed 32,306, possible 28,061.

6. The contract gate

`contracts/c360_customer.odcs.yaml` is built from the organisers' example DC-COLL-001 (owner, refresh, quality rules, allowed and not-allowed uses) and extended to ODCS v1.0.0. Publishing is blocked unless:

Rule	Status	Detail
Q1-unique-ids	**PASS**	1,052,300 golden IDs unique; every active source key maps to one

Rule	Status	Detail
Q2-linkage	**PASS**	98.66% of account keys linked; every open case resolves
Q3-match-safety	**PASS**	every auto link scores >= 0.95; no cluster holds two national IDs
Q3-review-precision	**PASS**	precision 99.3% on 300 reviewed predicted links; 300 of 300 pairs reviewed
Q4-contact-safety	**PASS**	0 violations
Q5-freshness	**PASS**	published 0.1h after the source extract
Q5-row-count	**PASS**	1,052,300 rows, +0.0% vs the last load
hardship-text	**PASS**	785,594 extractions; 11,464 golden IDs rest on them

7. Text, features, monitoring

- **Redaction first, always.** All 760,594 notes and 25,000 transcripts go through Presidio (English + French spaCy, Canadian SIN and postal codes, mangled cards and dates of birth). Full scan after: **0** leftovers. The LLM gateway won't even accept un-redacted text.
- **Labels: public only.** Text models are tuned and tested on the release's 500 public labels per source (30% held out). The rest are the organisers'.
- **Parity:** PASS: 0 of 56,000 cells differ (0.0000%; the limit is 0.1%).
- **Drift:** 10 features drift, 5 to watch, 11 stable (PSI vs training); 3 payroll features never appear in training rows.
- **Missing stays missing.** Transactions only cover a sample of customers and 13 months; outside that, those features are null, not 0.

8. What we can't measure

- **True match accuracy.** No ground truth; §5's human review and the exact-rule cross-check are the evidence.
- **Monthly roll rates.** The account snapshot only has quarter-ends (Dec 2023–Jun 2026), so roll rates are quarterly.
- **Generation rules.** The README says some fields were set by rule; we take them as given.

9. If we owned the source systems

1. Put one customer ID (or the CIF) in CRM: 892,088 of our links lean on a number only deposits and the bureau carry.
2. One date-of-birth format on the entry screen: 12.7% of CRM DOBs are ambiguous today.
3. Stop masking card phone numbers in the extract (58% are useless for matching).
4. Ship schema changes (`channel_v2`, the overdraft columns) in the dictionary first.
5. Keep a 300-link human review per release; it caught the only two bad links.

Rebuild: `uv run kept gate && uv run python scripts/quality_report_full.py` (with `kept serve` stopped).